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CHAPTER 1: EXECUTIVE SUMMARY & JOURNEY GENESIS

This document presents the key insights gained from four interconnected projects, tracing a journey from data exploration to building a reusable and reliable analytical system. By combining knowledge of Computer Science and Finance, the framework addresses a common challenge in financial analytics—replacing fragile, manual spreadsheet workflows with structured and scalable data pipelines. Rather than treating machine learning and text analytics as separate projects, each stage builds on the limitations of the previous one, creating an integrated framework where quantitative market data and qualitative business insights work together to support better decision-making.

CHAPTER 2: THE CONNECTED RESEARCH ARCHITECTURE

Instead of treating each project as a separate exercise, this framework follows a Scrum-inspired approach where larger objectives are divided into connected stages. Each stage builds on the previous one, creating a structured and logical progression throughout the project.



CHAPTER 3: THE UNIFIED RESEARCH QUESTION

"Can machine learning, natural language processing, and modern data engineering be combined into a reusable analytical framework that continues to deliver reliable insights as financial data grows over time?"

CHAPTER 4: MAJOR FINDINGS & EMPIRICAL EVIDENCE

FINDING 1: INCREASING MODEL COMPLEXITY DOES NOT ALWAYS IMPROVE BUSINESS UNDERSTANDING

Research Question: Does expanding model depth and ensemble complexity yield better practical value or superior risk management profiles on unseen market data?

OUT-OF-SAMPLE EVALUATION PARAMETER	BASELINE LOGISTIC MODEL	UNRESTRICTED ENSEMBLE
Training Metric Optimization	Balanced Baseline Capture	Maximum Optimization (>90%)
Out-of-Sample Performance Generalization	Stable (86.35% Accuracy)	Stagnates (82.58% Accuracy)
High-Confidence Correct Predictions (Unseen)	Cautious (73.1% to 70.0%)	Sharp Drop (80.2% to 73.8%)
High-Confidence Wrong Predictions (Risk Exposure)	Optimized to 5.0%	Aggressive Shift to 7.0%

Increasing model complexity does not always improve performance. In Phase 1, complex ensemble models achieved over 90% training accuracy, but they learned random patterns instead of meaningful market signals. As a result, they made more high-confidence mistakes on new data (7.0%). In comparison, the simpler Logistic Classification model adapted more cautiously to unseen data, resulting in fewer high-confidence errors (5.0%) and more reliable real-world performance.

PRACTICAL LEARNING
When choosing a model for a real-world analytics system, it is important to consider not only accuracy but also reliability. Simpler, well-calibrated models are often more dependable and safer for automated use than complex models that are prone to overfitting.

» Key Takeaway: Algorithmic simplicity often creates safer, more reliable corporate analytical solutions than unnecessary structural complexity.

FINDING 2: BUSINESS QUESTIONS SHOULD DRIVE ANALYTICAL DESIGN

Research Question: Why should a financial market prediction problem be reframed through direction-based classification rather than raw value-based magnitude?

MODELING DIMENSION	CONTINUOUS VALUE REGRESSION	DIRECTIONAL SUPERVISED CLASSIFICATION
Target Variable Structure	Exact daily return magnitudes	Binary directional movements (Up / Down)
Noise & Volatility Sensitivity	High; error residuals cluster heavily around volatility shocks	Minimal; filters out daily market noise
Out-of-Sample Performance	Test R-Squared drops from 86% to 82% under shocks	Maintained exactly at 86.35% accuracy

At first, the project used regression models to predict the exact market movement, but these models were heavily affected by market volatility, especially during events like COVID-19. The problem was then changed to predicting only the market direction (Up or Down) using a binary classification approach. This reduced the impact of daily price fluctuations and helped the model maintain a stable out-of-sample accuracy of 86.35% across different market conditions.

PRACTICAL LEARNING
The strategic framing of the target variable is more critical to a data project's success than the actual model choice. Aligning data definitions with a clear, directional business choice yields significantly higher operational reliability.

» Key Takeaway: Defining the business problem correctly is one of the most important skills in data science and has a major impact on the quality of the final solution.

FINDING 3: THEMES PROVIDE STRONGER BUSINESS SIGNALS THAN INDIVIDUAL ENTITIES

Research Question: How does the granularity of textual parsing affect the stability of an NLP signal?

NLP SIGNAL LAYER	DIRECT SENTIMENT (VADER)	CONTEXTUAL THEMES (SPACY)
Primary Analytical Unit	Isolated daily text tone	Grouped noun-chunk phrases
Resilience to Data Scarcity	Low (Highly skewed)	High (Isolates core narratives)
Market Directional Alignment	Stagnates at 50.75% Match	Optimized to 72.37% Match
Linguistic Noise Filtering	Susceptible to corporate bias	Filters out supporting entities

Using only sentiment scores from news articles did not produce a useful market signal, achieving an accuracy of just 50.75%, which was close to random chance. This happened because traditional sentiment analysis mainly measures whether the language is positive, negative, or neutral without understanding the actual business context behind the news. As a result, routine corporate announcements, promotional language, and genuinely important business events often received similar sentiment scores, making it difficult to identify information that could influence the market. To improve the analysis, the project used spaCy Named Entity Recognition (NER) to extract important words and phrases and grouped them into broader themes such as Risk & Uncertainty and Economic Growth. This shifted the focus from individual keywords to overall business narratives, reducing the influence of positive corporate language and other unnecessary text while creating more stable qualitative features. By understanding broader themes instead of isolated words, the system was better able to identify meaningful business events, capture important market conditions, and generate insights that were more relevant for financial analysis.

PRACTICAL LEARNING

To generate meaningful business insights from unstructured text, individual words and phrases should be grouped into broader themes instead of being analyzed separately. Organizing text into high-level categories creates more stable qualitative features and helps the system better understand the overall business context.

» Key Takeaway: Contextual narratives consistently outperform isolated data points when interpreting qualitative market behavior.

FINDING 4: REUSABLE SYSTEMS CREATE GREATER LONG-TERM VALUE THAN ONE-TIME ANALYSIS

Research Question: Can an analytical database infrastructure be engineered to absorb dynamic data updates automatically while eliminating manual spreadsheet-style errors?

OPERATIONAL PARAMETER	TRADITIONAL SPREADSHEET MODELING	LAYERED SQL VIEW ARCHITECTURE
Ingestion Mechanism	Manual data entry and formatting	Automated scripts feeding core tables
Logic Layer Separation	Faceted and fused into static cells	Decoupled via downstream SQL views
Calculative Maintenance	High risk of cell-shifting and breaks	Enforced datatype and schema safety

Traditional reporting systems often use static tables or spreadsheets, which can break when new data is added or calculations change. This project solved that problem by separating the data storage layer from the analysis layer in Google BigQuery. All predictions and calculations were built using dynamic SQL Views, allowing the system to update automatically whenever new data was added. When six months of additional financial data increased the dataset from 2,467 to 2,586 records, the entire pipeline updated successfully without requiring manual changes. Because calculations were performed at execution time instead of being stored in fixed cells, the system remained accurate, scalable, and easy to maintain.

PRACTICAL LEARNING

Data analytics systems should be designed for reuse from the beginning. Using a layered view architecture keeps calculations consistent and allows new data to flow through the system automatically without disrupting the existing workflow.

» Key Takeaway: Building systems that are reliable, reusable, and easy to maintain is just as important as developing accurate predictive models.

FINDING 5: VALIDATION IS AS IMPORTANT AS DEVELOPMENT

Research Question: Does a reusable analytics infrastructure preserve its original interpretation logic and distribution profiles when subjected to unseen data stress tests?

VALIDATION PARAMETER	PRE-APPEND BASELINE PROFILE	POST-APPEND STRESS TEST
Model Training State	Trained on stable historical data	Evaluated completely without retraining
Classification Accuracy	86.30% Baseline Accuracy	Preserved at exactly 86.35%
Distribution Stability	Initial signal baseline profile	Maintained baseline ranges under shifts

The true strength of an analytical system is not measured by how well it performs on historical data, but by how consistently it performs on new, unseen data over time. To evaluate this, six months of completely new out-of-sample data were added to the existing framework without changing the model structure. The Logistic Classification model maintained its strong accuracy of 86.35%, while the performance of the other models also remained close to their original baseline results. This consistency showed that the models had learned stable and repeatable market patterns instead of simply memorizing the historical dataset. The successful validation also demonstrated that the overall framework was reliable, stable, and capable of adapting to new data, making it more suitable for real-world financial analysis and long-term deployment.

PRACTICAL LEARNING

Data analytics systems should always be tested on new, unseen data over time. This helps confirm that the models have learned real and repeatable patterns instead of simply memorizing the data they were originally trained on.

» Key Takeaway: Analytical reliability is confirmed through testing against new, unseen data, not just training performance.

FINDING 6: DOCUMENTATION CONVERTS ANALYSIS INTO REUSABLE KNOWLEDGE

Research Question: How can complex, multi-environment technical experiments be converted into modular, explainable geographical assets?

OPERATIONAL DIMENSION	ISOLATED TECHNICAL CODE SCRIPTS	STANDARDIZED REUSABLE DESIGN
Code Explainability	Dense code comments; isolated logic	Modular notebooks and structured SQL Views
Workflow Onboarding	High friction; requires original author	Clear view-based checkpoints for pick-ups
System Visibility	Fragmented folders obscure calculations	Connected logical views provide direct map

Technical scripts and database queries can become difficult to understand and maintain if the underlying logic is not clearly documented. Without proper documentation, even well-written code can become hard to follow, making future updates and troubleshooting more challenging. In this project, clear documentation was used to connect the Python data processing pipeline with the Google BigQuery SQL views, ensuring that every step of the workflow was easy to understand. Queries, files, datasets, and processing stages were organized in a logical and structured manner, making the entire system transparent, consistent, and audit-ready. This approach also made the workflow easier to maintain, as new data or changes could be incorporated without disrupting the existing structure. As a result, other analysts can easily review, reproduce, validate, or expand the project without depending on the original developer, transforming the technical implementation into a reusable and valuable organizational asset.

PRACTICAL LEARNING

A data system is more valuable when its business logic is clearly documented and easy to understand. Organizing the workflow into well-defined, view-based stages makes the system easier to explain, maintain, scale, and reuse across different teams.

» **Key Takeaway:** Clear and standardized documentation helps turn a one-time technical project into a reliable system that can be used, maintained, and improved over the long term.

FINDING 7: BUSINESS INTERPRETATION CREATES REAL ANALYTICAL VALUE

Research Question: Where is the true commercial value of an AI/ML project located?

HTML ANALYSIS STRATEGY	RAW ALGORITHMIC METRIC OPTIMIZATION	INTEGRATED STRUCTURAL INTERPRETATION
Stream Focus	Chasing minor training error metrics	Analyzing the "Why" behind error profiles
Synergy	Models function as fragmented operations	Optimized Stock Match to 72.37%
Corporate Utility	Produces numbers detached from context	Converts statistics into business decisions

High model accuracy alone has limited value if the results cannot be explained in a business context. The real strength of this framework came from combining quantitative machine learning with qualitative text analytics. While individual signals, such as raw sentiment, produced weak results on their own, combining company earnings sentiment with broader market themes improved stock-level prediction accuracy to 72.37%. This integration connected technical data science methods with practical financial decision-making, showing that the greatest business value comes from turning analytical results into meaningful strategic insights.

PRACTICAL LEARNING

Data professionals should focus on how their work supports business decisions, not just on improving model accuracy. Real value comes from understanding how data, models, and business context work together to help leaders make better strategic decisions.

» **Key Takeaway:** Business interpretation is the final step that turns analytical results into practical insights that support better business decisions.

CHAPTER 5: CROSS-PROJECT STRUCTURAL PATTERNS

Across all four project stages, three key patterns consistently emerged.

First, **Simpler models proved more reliable**. Logistic Classification performed better than more complex models during unseen test periods because it focused on stable market patterns instead of temporary noise. This made it more dependable for real-world use.

Second, **Well-structured data improved results**. Changing the prediction target from exact price changes to market direction reduced the impact of volatility. In the NLP project, grouping related words into broader themes helped reduce bias and produced more meaningful insights.

Third, **A dynamic system design made the projects scalable**. Using Google BigQuery views instead of static spreadsheets allowed all calculations to update automatically whenever new data was added. This reduced manual work and improved accuracy.

CHAPTER 6: PROFESSIONAL GROWTH & REFLECTION

This project reflects a progression from simply writing code and building predictive models to designing complete analytical systems that can support real-world business decisions. Throughout the project, practical challenges such as limited historical data, biased financial text, changing market conditions, and system scalability highlighted that successful data science is not only about developing accurate models, but also about building solutions that are reliable, maintainable, and easy to use over time. These experiences strengthened a systems-oriented approach, where data pipelines, model performance, and business objectives work together as a single integrated framework.

The project also demonstrates the value of combining knowledge from Computer Science and Corporate Finance. Accurate models alone are not enough to create business value; analytical results must also be explained in a way that supports strategic decision-making. By integrating machine learning, natural language processing, financial analysis, and structured data engineering, the project transformed technical outputs into meaningful business insights. This multidisciplinary approach provides a strong foundation for developing practical, scalable, and decision-focused analytical solutions in real-world organizations.